

Austin Senna Wijaya

646-295-5341 | asw2215@columbia.edu | linkedin.com/in/austin-senna | github.com/Austin-Senna | austin-senna.github.io

EDUCATION

Columbia University

Bachelor of Science: Computer Science

New York, NY

Aug 2024 - May 2028

- GPA: 4.22/4.0; Full-Tuition Scholarship; Dean's List
- Relevant Coursework: Systems Programming, Data Structures, Databases, Natural Language Processing, Machine Learning.

EXPERIENCE

Software Engineer Intern

Jun 2026 – Aug 2026

Garner Health

New York, NY

- Engineered observability stack for newly launched B2B web portal, integrating **Datadog**, **OpenTelemetry**, and **Grafana** to establish end-to-end distributed tracing across frontend and backend services, surfacing previously invisible user-impacting bugs and increasing the team's bug resolution throughput by 3x.
- Automated telemetry provisioning, alerting, and monitoring infrastructure across Grafana and Datadog using **Terraform**, **GitLab CI/CD**, and **Helm**, replacing manual observability setup with reusable, version-controlled configuration.
- Deployed a self-improving autonomous on-call triage agent on **Kubernetes** using **Claude Agent SDK** with MCP tool access (PostgreSQL, Playwright, Jira, GitLab) to generate proactive alerts from telemetry data, investigate incident root cause, and ship fix pull requests; autonomously resolved **15+ production incidents**.

AI Research Assistant

Dec 2025 – Present

Columbia Data, Agents, and Processes Lab

New York, NY

- Engineered a high-throughput benchmark evaluation harness in **Strands SDK**, using parallel sandboxed **DuckDB/Python** runtimes for multi-agent question-answering tasks incorporating skills, hooks, and context compaction.
- Processed **9.5 TB / 40M docs** into an S3 data lake; built scalable **BM25** and **vector search** in LanceDB for massive-scale data retrieval for agents.
- Architected telemetry to capture tool-call traces and track agent reasoning to identify bottlenecks in agent framework, improving benchmark accuracy for GPT-5-mini from **5.16%** to **56.3%**.

Data Engineering Intern

Jun 2025 – Aug 2025

Ruangguru (Indonesia's largest edtech platform)

Jakarta, Indonesia

- Built scalable ETL pipelines in **BigQuery** to transform millions of telemetry events into structured session tables, mapping end-to-end payment conversion funnel to pinpoint drop-off points.
- Architected an automated web scraping, data ingestion, and sentiment analysis service utilizing **Puppeteer** and **Hugging Face** APIs, surfacing location review intelligence on live **Looker Studio** dashboards.

PROJECTS

ResearcherX: Local-First AI Research IDE

[Website](#)

- Led development of a local-first AI research editor (**Next.js**, **FastAPI**) using **GraphRAG** (LanceDB, NetworkX) to flag unsupported claims, contradictions, and citation drift in drafts; 2nd Place in Columbia's flagship startup competition.
- Engineered a real-time contradiction-detecting engine with minimized overhead via batched LLM calls and paragraph-hash caching, using **Pydantic**-validated output schemas to surface conflicts as inline, source-linked annotations.

TECHNICAL SKILLS

- **Languages:** Python, SQL, JavaScript / TypeScript, Java, C
- **AI & Data:** Strands Agents, Claude Agent SDK, Langchain, Pandas, PyTorch, Playwright, BeautifulSoup
- **Cloud & Infra:** AWS (EC2, S3, Bedrock), GCP (BigQuery, GKE), Grafana, OpenTelemetry, Kubernetes, Terraform, Docker
- **Framework & Tools:** Next.js, React, FastAPI, Litestar, Electron, GitHub, Claude Code

SELECTED PUBLICATIONS

- [1] *LakeQA: A Benchmark for Complex Exploratory QA over a Million-Scale Data Lake*, **ICML 2026**: Fourth author.
- [2] *SANA: What Matters for QA Agents over Massive Data Lakes?* **VLDB DASHSys 2026**: First author.
- [3] *MINA: Linear Probes Reveal Coding-Sequence Family Signal in Frozen DNA Encoders*, **PMLR MLCB 2026**: First author.

HONORS

- **Databricks Grounded Reasoning Cup 2026**: Columbia representative; worked with Google.
- **Jane Street Mystery Planet Trading Competition 2026**: 1st Place.
- **Hack@Brown 2026**: Finalist; Strongest Product Thinking award; worked with Google Ventures and Partiful.
- **Tencent Kepler Plan Competition 2025**: 2nd Place; Best Code; Best Product Idea; Popularity Award.
- **55th International Chemistry Olympiad 2023**: Final stage, 10th rank for Indonesian national team selection.